Compute-Constrained Road Damage Detection across Geographic and Sensor Domains

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***Abstract*—Road damage detection systems are commonly evaluated using pooled benchmarks, although deployment conditions vary across countries, sensors, resolutions, and road environments. This paper studies multidomain road damage detection on RDD2022 under a reproducible CPU-only budget. A leakage-aware protocol groups nearby frames, preserves negative images, and yields 38,385 labeled images with 55,006 valid instances. A 2.59-million-parameter YOLO11n detector is fine-tuned for 10 epochs on an equal-domain subset of 2,800 images in 28.5 minutes. On the held-out 3,703-image test split, it obtains 3.26% COCO mAP and 8.51% AP50 at 43.4 images/s. Domain mAP ranges from 0.42% for Norway to 6.11% for the United States, while small-object AP is only 0.26%. At a 0.25 score threshold, 0.92% of negative images produce a false detection. These results quantify the sharp accuracy cost of an extreme CPU and data budget and show why pooled accuracy alone is insufficient for assessing deployment reliability.**

***Index Terms*—road damage detection, object detection, compute-constrained learning, multi-domain evaluation, computer vision, RDD2022**

I. Introduction

Timely road-condition assessment is necessary for maintenance prioritization, yet manual surveys are costly to repeat at network scale. Images collected with smartphones and other widely available cameras provide a lower-cost sensing alternative, and modern object detectors can localize multiple damage types in a single pass [1]. Deployment reliability, however, depends on more than aggregate benchmark accuracy. Geographic appearance, pavement and marking standards, weather, acquisition platform, viewpoint, and image resolution may change simultaneously between training and operation.

RDD2022 provides a useful setting for studying this problem because it combines road imagery from seven acquisition domains spanning six countries [2], [3]. However, an aggregate score can conceal failure on a particular geography or sensor. Furthermore, a random image-level partition may place nearby frames from the same acquisition sequence in both training and evaluation sets, producing an optimistic estimate of deployment performance. Domain-dependent differences in class frequency, damage scale, and negative-image prevalence further complicate the comparison.

We address three research questions: *RQ1*, what detection quality is attainable with a compact model under a fully specified CPU-only budget; *RQ2*, how strongly does performance vary across the seven geographic and acquisition domains after equal-domain training; and *RQ3*, which classes, object scales, and negative-image conditions dominate the remaining errors?

This work therefore focuses on transparent compute constraints and domain-aware evaluation rather than only aggregate detection accuracy. Its contributions are:

- an audited cleaning pipeline and a deterministic, sequence-blocked partition of RDD2022 in COCO and YOLO formats;
- a deterministic CPU-budgeted baseline trained on an equal-domain subset with domain-specific negative prevalence preserved;
- domain-, class-, scale-, and negative-image-aware analysis rather than reliance on a single pooled score;
- a reproducible account of detection accuracy, latency, memory, and the accuracy cost imposed by compact CPU-only training.

II. Related Work

A. Road Damage Detection Benchmarks

Maeda et al. established an early large-scale benchmark for smartphone-based road inspection, releasing 9,053 images with 15,435 annotated damage instances and comparing convolutional object detectors for both server and mobile execution [1]. RDD2020 subsequently expanded the task from Japan to India and the Czech Republic, standardized four damage categories, and provided 26,336 images for multinational evaluation [4]. A companion study evaluated multiple combinations of country-specific training and test data and reported substantial degradation when models trained on Japanese roads were transferred to other countries [5].

RDD2022 extends this lineage to 47,420 images from six countries and introduces additional acquisition shifts, including motorbike, drone, high-resolution camera, and street-view imagery [3]. The associated CRDDC'2022 attracted solutions dominated by YOLO, Faster R-CNN, and detector ensembles; the best combined submission achieved an F1 score of approximately 76% [6]. These results demonstrate the value of model and data diversity, but leaderboard performance on the provided test sets does not by itself quantify domain-specific reliability under a fixed low-compute training procedure.

B. Detection Architectures

Dense one-stage detectors offer a practical accuracy–cost trade-off for road inspection. RetinaNet introduced focal loss to reduce the influence of easy background examples under extreme foreground–background imbalance [7]. Two-stage Faster R-CNN variants instead separate region proposal from region classification and retain an explicit objectness loss. The YOLO family emphasizes single-pass prediction and compact deployment. We use YOLO11n through the versioned Ultralytics implementation [8]; its 2.6 million parameters make repeated CPU optimization practical, while the low-resolution Faster R-CNN MobileNetV3 variant serves as an engineering qualification control. This choice targets a reproducible resource regime, not a state-of-the-art leaderboard claim.

C. Cross-Domain Robustness

Cross-country road damage detection has been studied through both pooled multinational training and domain adaptation. DA-RDD, for example, aligns image- and instance-level features between a labeled source and an unlabeled target domain [9]. This setting differs from domain generalization: adaptation uses target-domain samples during training, whereas generalization requires deployment to a domain that was unavailable during model fitting and selection. More broadly, Domain Generalised Faster R-CNN regularizes both classification and localization under domain shift [10]. DomainBed further shows that fair conclusions depend on consistent architectures, tuning budgets, and model-selection rules, and that a carefully implemented empirical-risk-minimization baseline can be difficult to outperform [11].

Following these lessons, our study establishes a controlled empirical-risk minimization baseline under a fixed CPU budget. Its distinguishing focus is not a new adaptation loss, but a sequence-aware, equal-domain assessment that preserves negative images and reports domain-level rather than only pooled performance. We deliberately avoid interpreting domain slices as unseen-domain generalization.

III. Problem Formulation

A. Multi-Domain Object Detection

Let $\mathcal{D} = \{1, \ldots, D\}$ denote the acquisition domains, with $D = 7$, and let

$$\mathcal{Z}_d = \{(x_{di}, Y_{di})\}_{i=1}^{n_d}, \quad Y_{di} = \{(b_{dij}, c_{dij})\}_{j=1}^{m_{di}}, \tag{1}$$

where $x_{di} \in \mathbb{R}^{H_i \times W_i \times 3}$, normalized box $b_{dij} \in [0,1]^4$, and $c_{dij} \in \mathcal{C} = \{\mathrm{D00}, \mathrm{D10}, \mathrm{D20}, \mathrm{D40}\}$. A negative image is represented without a synthetic background box, i.e., $Y_{di} = \varnothing$ and $m_{di} = 0$. A detector maps an image to K scored hypotheses,

$$f_\theta(x) = \{(\hat{b}_k, \hat{\boldsymbol{p}}_k)\}_{k=1}^{K}, \qquad \hat{\boldsymbol{p}}_k \in [0,1]^{|\mathcal{C}|}. \tag{2}$$

Let $\mathcal{A}_i$ be the multiscale candidate locations for image i. For candidate a and target j, the task-alignment score is

$$t_{iaj} = p_{ia,c_j}^{\alpha}\, \mathrm{IoU}(\hat{b}_{ia}, b_{ij})^{\beta}, \tag{3}$$

and $\mathcal{P}_i$ contains the top-k candidates for each target whose centers fall inside that target. The selected alignment values are normalized into soft class targets $s_{iac} \in [0,1]$. For Bernoulli cross-entropy $\mathrm{BCE}(p,s) = -s \log p - (1-s)\log(1-p)$, classification is

$$L_{\mathrm{cls}} = \sum_i \sum_{a \in \mathcal{A}_i} \sum_{c \in \mathcal{C}} \mathrm{BCE}(p_{iac}, s_{iac}). \tag{4}$$

Thus, when $Y_i = \varnothing$, every $s_{iac} = 0$: localization vanishes but background classification remains active.

For a continuous target distance $y \in [0, R-1)$ and categorical distribution $q \in \Delta^{R-1}$, let $l = \lfloor y \rfloor$ and $u = \min(l+1, R-1)$. The distribution focal loss is

$$\mathrm{DFL}(q,y) = -(u-y)\log q_l - (y-l)\log q_u. \tag{5}$$

With $Z = \max(\sum_{i,a,c} s_{iac}, 1)$ and matched target $\mu(i,a)$, the frozen YOLO objective is represented by

$$\begin{aligned}\mathcal{L}(\theta) = {} & \frac{\lambda_{\mathrm{cls}}}{B} L_{\mathrm{cls}} \\ & + \frac{1}{Z}\sum_i \sum_{a \in \mathcal{P}_i} s_{ia,c_{\mu(i,a)}} \Big[\lambda_{\mathrm{box}} \\ & \cdot \big(1 - \mathrm{CIoU}(\hat{b}_{ia}, b_{i\mu(i,a)})\big) + \lambda_{\mathrm{dfl}} \sum_{r \in \{l,t,r,b\}} \mathrm{DFL}(q_{iar}, y_{iar}) \Big],\end{aligned} \tag{6}$$

where $(\lambda_{\mathrm{box}}, \lambda_{\mathrm{cls}}, \lambda_{\mathrm{dfl}}) = (7.5, 0.5, 1.5)$ and B is batch size. This exposes how classification, geometry, and discretized boundary regression compete under a fixed optimization budget.

B. Domain Risk and Negative Prevalence

For loss ℓ, the empirical domain risk and a weighted source risk are

$$\widehat{R}_d(\theta) = \frac{1}{n_d}\sum_{i=1}^{n_d} \ell(f_\theta(x_{di}), Y_{di}), \qquad \widehat{R}_{\mathcal{S}}^{\boldsymbol{\alpha}}(\theta) = \sum_{d \in \mathcal{S}} \alpha_d \widehat{R}_d(\theta), \tag{7}$$

where $\alpha_d = n_d / \sum_{s \in \mathcal{S}} n_s$ recovers natural pooled sampling and $\alpha_d = 1/|\mathcal{S}|$ gives equal domain weight. If $q_d = n_d^{-1} \sum_i \mathbf{1}[m_{di} = 0]$, then

$$\widehat{R}_d = (1 - q_d)\widehat{R}_d^{+} + q_d \widehat{R}_d^{-}, \tag{8}$$

making domain-dependent negative prevalence an explicit change in the risk mixture rather than a secondary dataset statistic.

Let the CPU training budget be $B_{\mathrm{tr}} = 2800$ images and define domain quotas $B_d \in \{\lfloor B_{\mathrm{tr}}/D \rfloor, \lceil B_{\mathrm{tr}}/D \rceil\}$ such that $\sum_d B_d = B_{\mathrm{tr}}$. For binary selection z_{di} and negative prevalence q_d, the sampler obeys

$$\begin{aligned} & \sum_i z_{di} = B_d, \qquad \sum_i z_{di} \mathbf{1}[m_{di} = 0] = \mathrm{round}(q_d B_d), \\ & \widehat{R}_{\mathrm{bal}}(\theta) = \frac{1}{D}\sum_{d=1}^{D} \frac{1}{B_d} \sum_i z_{di} \ell(f_\theta(x_{di}), Y_{di}). \end{aligned} \tag{9}$$

The frozen run uses $B_d = 400$ for every domain and $E = 10$ epochs, hence $EB_{\mathrm{tr}} = 28{,}000$ image presentations. Besides pooled AP, we report

$$AP_{\mathrm{macro}} = D^{-1}\sum_{d=1}^{D} AP_d(\theta), \qquad AP_{\mathrm{worst}} = \min_d AP_d(\theta), \tag{10}$$

so a large domain cannot conceal failure on a smaller one. These are domain-sliced in-distribution results; they are not estimates of adaptation or unseen-domain generalization.

C. Constrained Sequence-Block Partition

Within each domain, let $\mathcal{G}$ be the sequence blocks and define the feature vector

$$v_g = (N_g, N_g^-, N_g^{\mathrm{D00}}, N_g^{\mathrm{D10}}, N_g^{\mathrm{D20}}, N_g^{\mathrm{D40}})^\top \in \mathbb{N}^6. \tag{11}$$

For split $r \in \{\mathrm{tr}, \mathrm{va}, \mathrm{te}\}$, binary assignment a_{gr}, target ratio $\rho = (0.8, 0.1, 0.1)$, total $V = \sum_g v_g$, and $w = (10, 2, 3, 3, 3, 3)$, set

$$A_{rk} = \sum_{g\in\mathcal{G}} a_{gr} v_{gk}, \qquad T_{rk} = \rho_r V_k, \tag{12}$$

$$\delta_{rk} = \frac{A_{rk} - T_{rk}}{\max(T_{rk}, 1)}, \qquad o_{rk} = \frac{[A_{rk} - 1.08T_{rk}]_+}{\max(T_{rk}, 1)}. \tag{13}$$

The implemented assignment approximately solves

$$\begin{aligned} \min_a \quad & J(a) = \sum_{r,k} w_k(\delta_{rk}^2 + 3o_{rk}^2), \\ \text{s.t.} \quad & a_{gr} \in \{0,1\}, \qquad \sum_r a_{gr} = 1. \end{aligned} \tag{14}$$

Rare blocks are greedily placed first using a seed-stable order, after which 30,000 seeded pairwise-swap proposals are accepted only when they decrease J. The constraint in (14) makes sequence leakage impossible by construction, while the asymmetric o_{rk} term discourages overshooting any target feature by more than 8%.

IV. Dataset and Exploratory Analysis

A. RDD2022

RDD2022 contains 47,420 road images collected in China, the Czech Republic, India, Japan, Norway, and the United States [2]. China is represented by two acquisition domains, drone and motorbike, resulting in seven domains overall. The official task uses four damage categories: longitudinal cracks (D00), transverse cracks (D10), alligator cracks (D20), and potholes (D40).

The released data contain 38,385 labeled training images and 9,035 unlabeled challenge-test images. We found 65,712 annotated objects before cleaning. Of these, 55,007 belong to the four official classes and 10,705 use additional labels. One official bounding box has zero width and is removed, leaving 55,006 valid target instances.

B. Class, Scale, and Domain Imbalance

Table I summarizes the cleaned official instances. D00 accounts for nearly half of the annotations, whereas D40 accounts for only 11.9%. The most frequent class is therefore approximately four times more common than the least frequent class. Moreover, 41.3% of valid boxes occupy less than 1% of their image area, indicating a substantial small-object detection challenge.

Negative-image prevalence also varies sharply by domain. No United States training image is negative under the four-class definition, compared with 64.3% of Norway, 62.1% of the Czech Republic, and 58.2% of India. This variation motivates domain-aware reporting and explicit negative sampling.

V. Data Preparation and Evaluation Protocol

A. Cleaning Rules

Only D00, D10, D20, and D40 are treated as detection targets. Additional labels are removed, while images left without an official target are retained as negative examples. Bounding boxes with non-positive area or coordinates outside the image are excluded rather than silently clipped. A single canonical annotation table is used to generate both COCO and YOLO artifacts. The original unlabeled challenge-test set is kept separate from all internal model selection and evaluation. Automated validation checks image existence, unique identifiers, box geometry, normalized YOLO coordinates, COCO references, split membership, and class mappings. It verified all 38,385 labeled images and 55,006 retained boxes without a validation failure.

B. Leakage-Aware Internal Split

Numeric image identifiers are grouped into blocks of 50 consecutive indices. All images from a block remain in the same subset, reducing leakage from nearby frames. With seed 2026, blocks are assigned independently within each domain toward an 80/10/10 train/validation/test target. The deterministic assignment minimizes a weighted discrepancy in image count, negative images, and instances of each class, followed by block swaps that improve the same objective without breaking groups. The resulting split contains 31,021 training, 3,661 validation, and 3,703 internal-test images. No sequence block occurs in more than one subset.

C. CPU-Budgeted Training Subset

The full validation and internal-test partitions are retained, but optimization uses a deterministic subset of 2,800 training images. Exactly 400 are sampled from each domain. Within domain d, positives and negatives are sampled separately so the realized negative count equals $\mathrm{round}(400q_d)$. The resulting subset contains 925 negative images and 4,019 instances: 2,047 D00, 985 D10, 626 D20, and 361 D40. Absolute image lists and realized counts are stored with seed 2026. Equal quotas implement the risk in (9) while retaining the substantial domain-specific difference in background prevalence.

Fig. 1. RDD2022 composition, official class frequency, class distribution by domain, and relative bounding-box area.

TABLE I
OFFICIAL RDD2022 CLASS DISTRIBUTION

Class	Damage type	Instances	Share
D00	Longitudinal crack	26,016	47.3%
D10	Transverse crack	11,830	21.5%
D20	Alligator crack	10,616	19.3%
D40	Pothole	6,544	11.9%

TABLE II
CLEAN INTERNAL SPLIT

Split	Images	D00	D10	D20	D40
Train	31,021	20,835	9,457	8,501	5,243
Validation	3,661	2,588	1,190	1,054	652
Test	3,703	2,593	1,183	1,061	649

D. Metrics

The primary metric is COCO average precision averaged over intersection-over- union thresholds from 0.50 to 0.95 [12]. We also report AP50, AP75, average recall, per-class AP, per-domain AP, and AP for small, medium, and large objects. Predictions are capped at 100 detections per image for COCO evaluation. Computational reporting includes latency, throughput, parameter count, training time, and peak memory measured on fixed hardware. Domain macro and worst-domain AP use the same shared prediction set. The run is deterministic at seed 2026, but it is not repeated across seeds; consequently, we report no unsupported standard deviation and treat training variance as a limitation.

VI. EXPERIMENTAL SETUP

A. Model Selection

YOLO11n is the primary compact detector. It has 2.59 million trainable parameters and is initialized from the official COCO checkpoint. Before the internal test was accessed, CPU qualification compared it with the official low-resolution Faster R-CNN MobileNetV3 implementation. Faster R-CNN learned from negative-only batches but required 2.24 s per two-image optimization batch; YOLO11n processed eight images in approximately 0.45 s. The former is retained as an engineering control rather than a commensurate scientific baseline.

B. Training and Model Selection

The frozen run uses 320×320 inputs, batch size 8, and 10 epochs, giving 28,000 image presentations. AdamW uses initial learning rate 0.00125, final factor 0.05, weight decay 5×10^{-4}, and one warm-up epoch. Augmentation comprises horizontal flipping, HSV perturbation, translation, scaling, and mosaic, with mosaic disabled for the final two epochs. There is no early stopping: the final-epoch checkpoint is evaluated on validation, and the internal test is evaluated once after all choices are frozen.

C. Implementation and Reproducibility

The run uses Python 3.14.7, PyTorch 2.10.0, TorchVision 0.25.0, and Ultralytics 8.3.203 on an Intel Core i5-10500T CPU with six physical cores, 31 GiB RAM, and no available CUDA device. Training artifacts record the seed, absolute sample lists, software versions, model arguments, losses, metrics, and checkpoint provenance. Integration diagnostics exercised negative-only batches, serialization, deterministic reruns, and official COCO evaluation; diagnostic values are excluded from the internal-test table.

VII. Results

Table III reports the single frozen evaluation. Training took 1,709.6 s (28.5 min). From the first to the final epoch, classification, box, and distribution-focal training losses decreased from 3.833 to 2.774, 2.998 to 2.300, and 2.363 to 1.905, respectively. Final validation mAP was 3.01%, close to the 3.26% test result, and no test value was used for model selection.

Scale-stratified AP was 0.26%, 2.71%, and 3.63% for small, medium, and large objects. AR_{100} reached 27.33%, showing that the detector often proposed a nearby candidate but localized or ranked it insufficiently for the stricter AP integral. By class, mAP/AP50 was 3.83/9.77% for D00, 3.01/9.17% for D10, 5.46/12.88% for D20, and 0.75/2.23% for D40. Thus, the rare pothole class was the weakest, whereas alligator cracks were the strongest despite being less frequent than D00.

Mean preprocessing, inference, and postprocessing time summed to 23.0 ms per image, or 43.4 images/s for the measured model stages. The complete evaluator, including file I/O and all pooled, class, and domain COCO computations, took 162.1 s and peaked at 1.33 GiB resident memory. Among 1,414 negative test images, 13 (0.92%) produced at least one detection at score 0.25; the mean was 0.0149 detections per negative image. The low false-positive rate therefore coexists with low recall and AP rather than establishing overall reliability.

VIII. Domain-Wise Analysis

All domain statistics are computed by slicing one shared internal-test prediction file. This controls inference settings and prevents domain-specific threshold tuning. We report pooled, macro, and worst-domain AP according to (10). Because every domain contributes training images, any difference reflects within-benchmark heterogeneity, sample composition, and acquisition difficulty jointly; it does not identify a causal geographic effect or unseen-domain generalization.

TABLE III
Frozen Internal-Test Result (percent except FPS)

Model	mAP	AP50	AP75	FPS
YOLO11n	3.26	8.51	1.82	43.4

Table IV exposes a 14.6-fold range between the United States and Norway. Domain-macro mAP is 3.47%, whereas pooled mAP is 3.26% and worst-domain mAP is 0.42%. Equal image quotas did not imply equal positive supervision: the sampler deliberately preserved each domain's negative-image rate, which is 64.3% for Norway but zero for the United States. Acquisition geometry, resolution, class mix, annotation practice, and test composition are also confounded with domain, so the table supports a disparity finding but not a causal geographic explanation.

IX. Ablation Studies

Compute qualification established two useful boundary cases. With a new four-class head, Faster R-CNN MobileNetV3 produced zero validation AP after 100 steps; at 500 steps it reached 1.25% mAP and 3.29% AP50 on a fixed 500-image validation subset. The equal-domain YOLO11n pilot used 1,400 images for five epochs and reached 1.75% mAP and 5.35% AP50 on all 3,661 validation images in 8.8 minutes of CPU training. These noncommensurate pilots informed model and budget selection only and are not treated as test-set comparisons. Higher resolution, tiling, removal of negatives, and repeated seeds remain future ablations.

X. Limitations and Discussion

The absolute AP values are not competitive with unconstrained RDD2022 systems. This is an expected but important boundary result: COCO pretraining cannot fully compensate for only 28,000 low-resolution image presentations. The near-monotonic loss reduction and similar validation/test AP suggest budget underfitting rather than test-specific collapse. Increasing resolution is the most direct hypothesis for the 0.26% small-object AP, but it would also raise CPU cost superlinearly and must be tested rather than assumed.

The current protocol evaluates only the shifts represented in RDD2022 and does not cover all pavement materials, climates, cameras, or maintenance taxonomies. Country and sensor are partly confounded—most domains use a characteristic acquisition setup—so a performance difference cannot be interpreted as a causal effect of geography alone. Sequence blocks are inferred from numeric identifiers because timestamps and trajectories are unavailable; grouping nearby identifiers reduces leakage but cannot guarantee removal of every near duplicate.

TABLE IV
DOMAIN-SLICED INTERNAL-TEST PERFORMANCE

Domain	Images	mAP (%)	AP50 (%)
China–drone	250	5.83	16.75
China–motorbike	200	6.07	15.35
Czech Republic	283	2.06	6.24
India	725	0.95	2.54
Japan	1,045	2.84	7.79
Norway	750	0.42	1.35
United States	450	6.11	15.11

Annotation tools and source-specific policies may also contribute to domain differences. Finally, the official challenge-test labels are not public, preventing independent evaluation on that split without an external service. The 320-pixel input is especially restrictive because 41.3% of boxes occupy less than 1% of the original image; resizing can erase fine crack structure. Equal domain quotas improve representation but depart from natural deployment prevalence, and only one deterministic training seed is available. Finally, all evaluated domains occur in training, so the study measures multidomain performance, not adaptation or generalization to an unseen domain. Claims are restricted accordingly.

XI. CONCLUSION

This study established an auditable CPU-only baseline for multidomain road damage detection. Sequence-blocked splitting, explicit negative preservation, and equal-domain sampling prevent common sources of leakage and silent domain imbalance. YOLO11n completed the frozen 10-epoch run in 28.5 minutes and processed the test set at 43.4 images/s, but achieved only 3.26% COCO mAP. The 14.6-fold domain range, 0.42% worst-domain mAP, 0.75% pothole AP, and 0.26% small-object AP show that computational feasibility does not imply deployment readiness. The resulting checkpoint is therefore a transparent lower-compute reference rather than a state-of-the-art claim. Future work should spend additional compute first on resolution-aware training or tiling, then test repeated seeds and domain-balanced positive supervision, while retaining the same untouched sequence-blocked evaluation protocol.

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